

Part 4: Comparing TCP and UDP

Task 1: Reliability & Connection

Feature	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented and requires a 3-way handshake (SYN, SYN-ACK, ACK) before data transfer. It guarantees reliable delivery via ACKs and retransmissions. UDP is connectionless and does not guarantee delivery.
Data Integrity and Ordering	TCP	TCP uses Sequence Numbers and Acknowledgment Numbers to reassemble packets in the correct order and detect missing data. UDP sends datagrams independently without ordering.

Task 2: Use Cases and Performance

Category	TCP	UDP
Use Cases	Web Browsing (HTTP/HTTPS), File Transfer (FTP), Email (SMTP), Remote Access (SSH).	Live Video Streaming, Voice Calls (VoIP), Online Gaming, DNS Queries.
Performance	Lower speed / higher overhead due to connection setup, ACKs, flow control, and error recovery.	Higher speed / low overhead with a small 8-byte header and no connection setup delays.